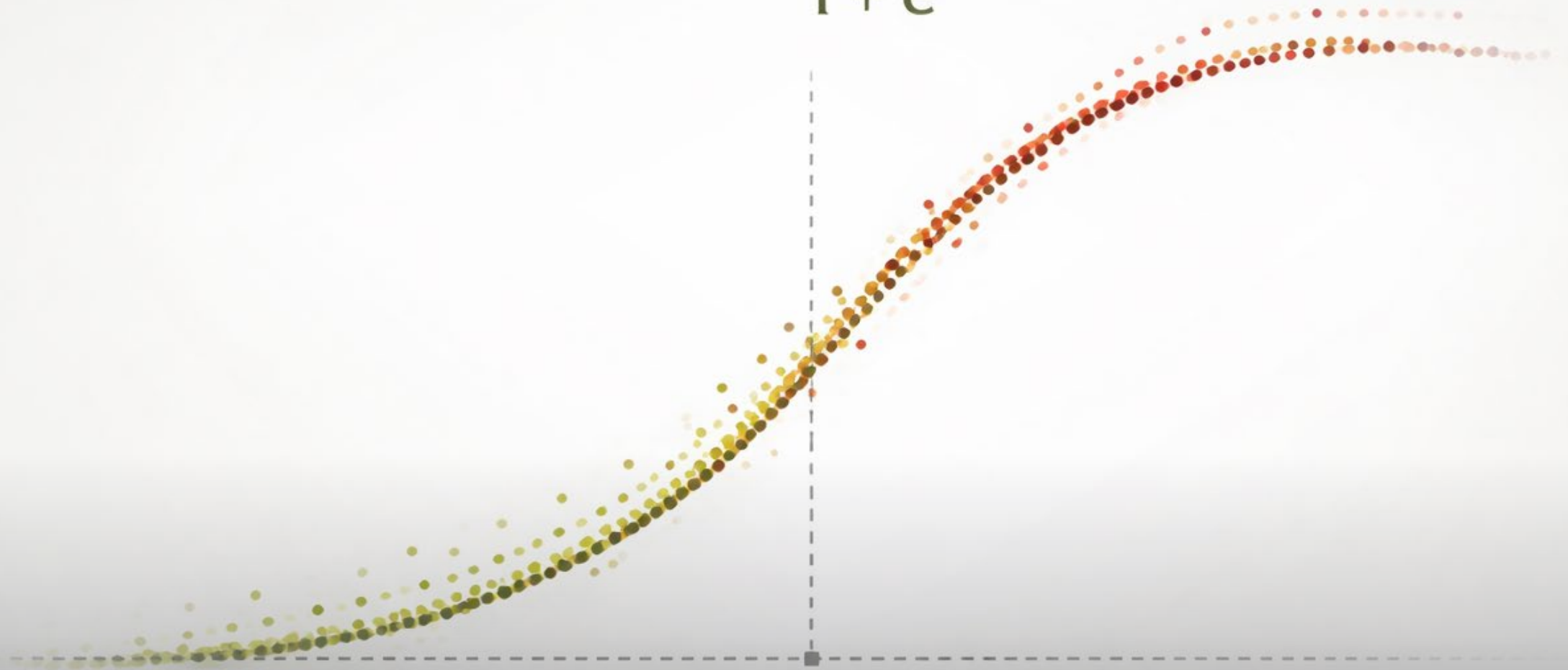




$$\sigma(z) \approx \frac{1}{1 + e^{-z}}$$



LOGISTIC REGRESSION



OUTLINE

One idea runs through the whole lecture: keep the linear score $z = w^T x + b$, and change what we do with it.

Part 1 — Setting Up the Model

How do we adapt linear regression for a (probabilistic) classifier?

Part 2 — Fitting the Model

How do we learn the parameters of this model? (w and b)

Part 3 — Using the Model

How do we turn a probability into a decision, and how do we check our work?



PART 1 — SETTING UP THE MODEL

From a binary outcome to a probability model.

- Why is linear regression the wrong tool here?

Unbounded predictions: $\hat{y} \in (-\infty, \infty)$, not $[0,1]$

Wrong error model: assumes Gaussian noise, not Bernoulli

- We need a function that maps a real-valued score into $(0,1)$?
-



MOTIVATION: BINARY OUTCOMES

Goal: predict a binary outcome $y \in \{0,1\}$ from a feature vector $x \in \mathbb{R}^p$

Running example: predicting credit-card default ($y = 1$) from account balance.

Notation used throughout

- $x = (x_1, \dots, x_p)$: features (predictors) measured on one observation
 - $w \in \mathbb{R}^p$: weights (coefficients), one per feature
 - $b \in \mathbb{R}$: bias (intercept)
 - $p = P(y = 1 \mid x)$: probability the outcome is 1, given the features
-

LINEAR VS. LOGISTIC, SIDE BY SIDE

Both panels fit default probability against account balance.

Left: the linear fit leaves the $[0,1]$ range, predicting negative probabilities at low balances. Right: the sigmoid fit stays in $(0,1)$ and bends where default risk actually changes.

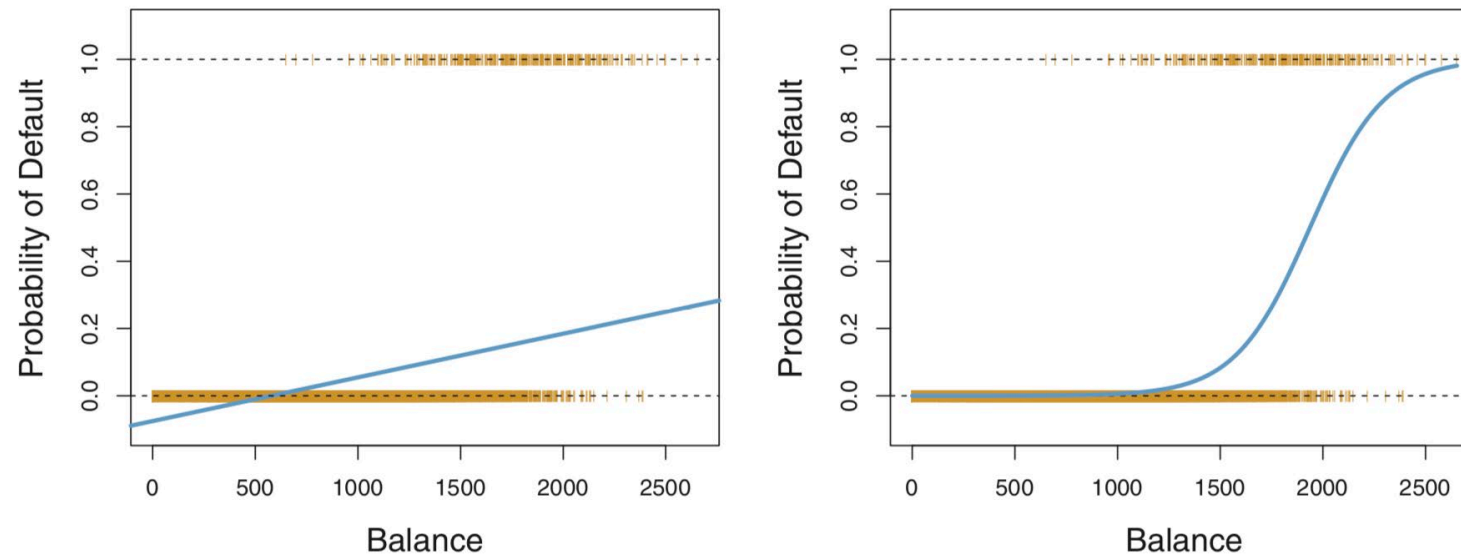


Figure from Introduction to Statistical Learning by James et al.

FROM LINEAR SCORES TO PROBABILITIES

Linear predictor (score):

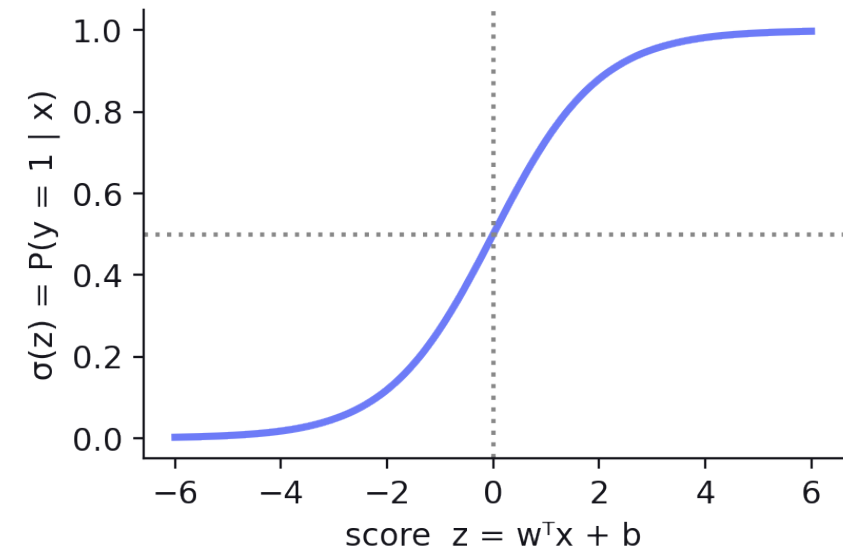
- $z = w^T x + b$
 - In linear regression, z is out output! (prediction)
 - Here, it is a **latent score**

We need a smooth, monotonic map from $\mathbb{R}$ into $(0,1)$:

- $\sigma(z) = 1 / (1 + e^{-z})$
- Sigmoid or Logistic function

Model:

- $P(y = 1 \mid x) = \sigma(w^T x + b)$
- What is $P(y = 0 \mid x)$?
- By symmetry: $P(y = 0 \mid x) = 1 - \sigma(z) = \sigma(-z)$





ODDS AND THE LOGIT

Odds

The ratio of success to failure probability:

$$\text{odds}(x) = p / (1 - p)$$

$$p \in (0,1) \Rightarrow \text{odds} \in (0, \infty)$$

A probability of 0.2 is odds of 0.25 — one default for every four non-defaults.

Odds rescale a probability onto $(0, \infty)$, but they are still not linear in the features.

Log-odds (the logit)

The sigmoid and the logit are exact inverses — each undoes the other:

$$p = \sigma(w^T x + b) \Leftrightarrow \text{logit}(p) = w^T x + b$$

where $\text{logit}(p) = \log(p / (1 - p))$

$$\text{and } \sigma(z) = 1 / (1 + e^{-z})$$

σ carries a log-odds score into a probability; the logit carries a probability back to the score.

Taking the log is what makes the scale linear — logistic regression is a linear model on the log-odds scale.



READING THE COEFFICIENTS

The log-odds is an ordinary linear equation:

$$\text{logit}(p) = \log(p / (1 - p)) = w^T x + b$$

So read w_j exactly as a coefficient in linear regression — the change in the response for a one-unit increase in x_j , holding the others fixed. The only difference is that the response is the log-odds.

Exponentiate to move to the odds scale:

$$\text{odds} = p / (1 - p) = \exp(w^T x + b)$$

A one-unit increase in x_j therefore multiplies the odds by $\exp(w_j)$ — the odds ratio.

Rules of thumb

- $w_j > 0$ raises the probability of $y = 1$; $w_j < 0$ lowers it; $w_j = 0$ means no effect
- $w_j = 0.7 \Rightarrow \exp(0.7) \approx 2$, so each unit of x_j roughly doubles the odds
- Default example: $w = 2.0$ for balance (in \$1,000s) $\Rightarrow \exp(2.0) \approx 7.4$, so each extra \$1,000 multiplies the odds of default by about 7.4

Note

Coefficients are not changes in probability — the same odds ratio moves a probability a lot near 0.5 and very little near 0 or 1.



PART 2 — FITTING THE MODEL

From a model to fitted values of w and b .

- What probability model generated the data?
 - What objective does that model imply?
 - How do we minimize it when there is no closed form?
-



THE BERNOULLI MODEL

Explicitly model the data-generating process:

$$y \mid x \sim \text{Bernoulli}(p), \quad p = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

A Bernoulli random variable equals 1 with probability p and 0 with probability $1 - p$ — a single weighted coin flip. Each customer either defaults or does not; the model sets the bias of that coin.

Why this choice

- It matches the binary nature of the data
- It gives a principled, likelihood-based





FROM LIKELIHOOD TO CROSS-ENTROPY

- Likelihood: how probable the observed data are under the model. For one observation:

$$P(y \mid x) = p^y (1 - p)^{1-y}$$

- The exponents just select the right factor: p when $y = 1$, and $1 - p$ when $y = 0$.

- **Log-likelihood for n independent observations:**

$$\ell(w, b) = \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

- **Loss = negative log-likelihood, also called cross-entropy loss:**

$$\mathcal{L}(w, b) = - \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

- Maximizing the likelihood is exactly minimizing cross-entropy. A customer who defaulted but was given $p \approx 0$ contributes a very large loss.
-



OPTIMIZATION I: THE PROBLEM

Objective:

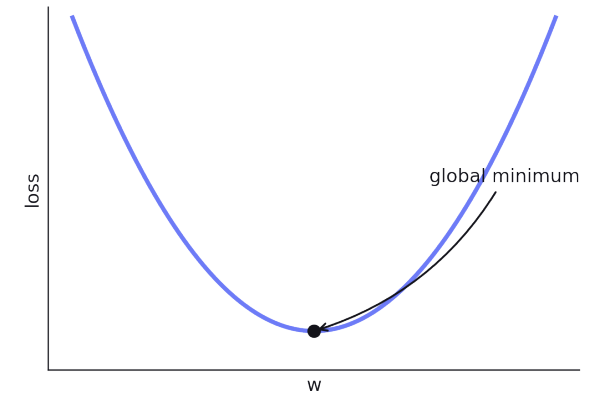
min over (w, b) : $\mathcal{L}(w, b)$

Properties

- Convex — the loss surface is one bowl, so any local minimum is the global one
- No closed form — there is no normal equation, so we solve numerically

Methods: gradient descent, stochastic gradient descent (SGD), Newton's method.

Can add an L1 or L2 penalty to this problem convex while controlling overfitting.



OPTIMIZATION SETUP



OPTIMIZATION II: GRADIENT DESCENT

Gradient of the loss with respect to the weights:

$$\partial \mathcal{L} / \partial w = \sum_i (p_i - y_i) x_i$$

Update rule: step downhill, $w \leftarrow w - \eta \cdot \partial \mathcal{L} / \partial w$, where η is the learning rate.

A direct parallel to linear regression

- Linear regression: updates use residuals ($\hat{y} - y$)
- Logistic regression: updates use probabilistic errors ($p - y$)

Interpretation

- $p > y$: the gradient pushes the prediction down toward 0
 - $p < y$: the gradient pulls the prediction up toward 1
 - $p \approx y$: the signal is small — that observation is already explained
-



PART 3 — USING THE MODEL

From a fitted probability to a decision.

- How does a probability become a predicted class?
 - Who decides the threshold, and on what grounds?
 - Can you run the whole pipeline yourself?
-



CLASSIFICATION: THE DECISION RULE

Logistic regression outputs probabilities. Classification needs one more ingredient — a decision threshold τ :

$$\hat{y} = 1 \text{ if } P(y = 1 \mid x) \geq \tau, \text{ otherwise } \hat{y} = 0$$

The standard choice is $\tau = 0.5$, which is equivalent to classifying on the sign of the score z .

Why $\tau = 0.5$ is only a default

- It implicitly assumes a false positive and a false negative cost the same.
 - Nothing in the fitting procedure chose it — the model never saw τ .
 - Changing τ changes the predictions without refitting the model.
-



CHOOSING THE THRESHOLD

Set τ from the cost of each error type, not by habit.

- Medical screening: a missed case (false negative) is far costlier than a false alarm, so lower τ — catch more positives and accept more false positives
- Fraud review: each flag costs analyst time, so raise τ to protect precision
- Credit default: a lender who loses more on a bad loan than on a declined application should lower τ

Types of mistakes

False positive: predicted 1, actually 0. False negative: predicted 0, actually 1.

Sweeping τ from 0 to 1 traces the ROC and precision–recall curves — the subject of the next lecture.



CHECK YOUR UNDERSTANDING

A model fit to the Default data (balance in \$1,000s) gives $w = 2.0$, $b = -7.0$. For a customer with a \$2,000 balance ($x = 2.0$), work out the predicted probability and the class label.

Step 1 — compute the score

$$z = w^T x + b = 2.0(2.0) - 7.0 = -3.0$$

Step 2 — convert the score to a probability

$$p = \sigma(-3.0) = 1 / (1 + \exp(3.0)) \approx 0.047$$

Step 3 — apply the decision rule

$$p = 0.047 < \tau = 0.5 \Rightarrow \text{predict } \hat{y} = 0 \text{ (no default)}$$

Now try it yourself: what is the predicted class at a \$4,000 balance, and what threshold would flip the \$2,000 answer?



CONCEPTUAL SUMMARY

1. Start with the linear regression structure: $z = w^T x + b$
2. Interpret z as the log-odds (logit) of the outcome
3. Apply the sigmoid to turn the score into a probability
4. Model the outcome with a Bernoulli likelihood
5. Minimize the cross-entropy loss (convex, no closed form)
6. Solve with gradient-based optimization
7. Apply a threshold τ to turn probabilities into class decisions

You should now be able to

- Compute a predicted probability from w , b , and x by hand
 - Interpret any coefficient as an odds ratio, and say why it is not a probability change
 - Justify a threshold from the relative cost of the two error types
-